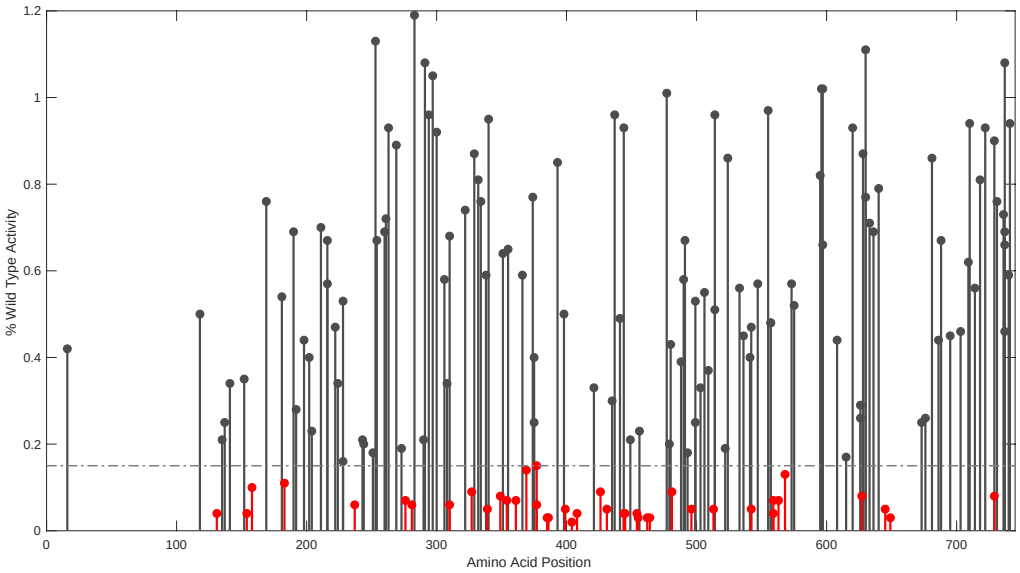
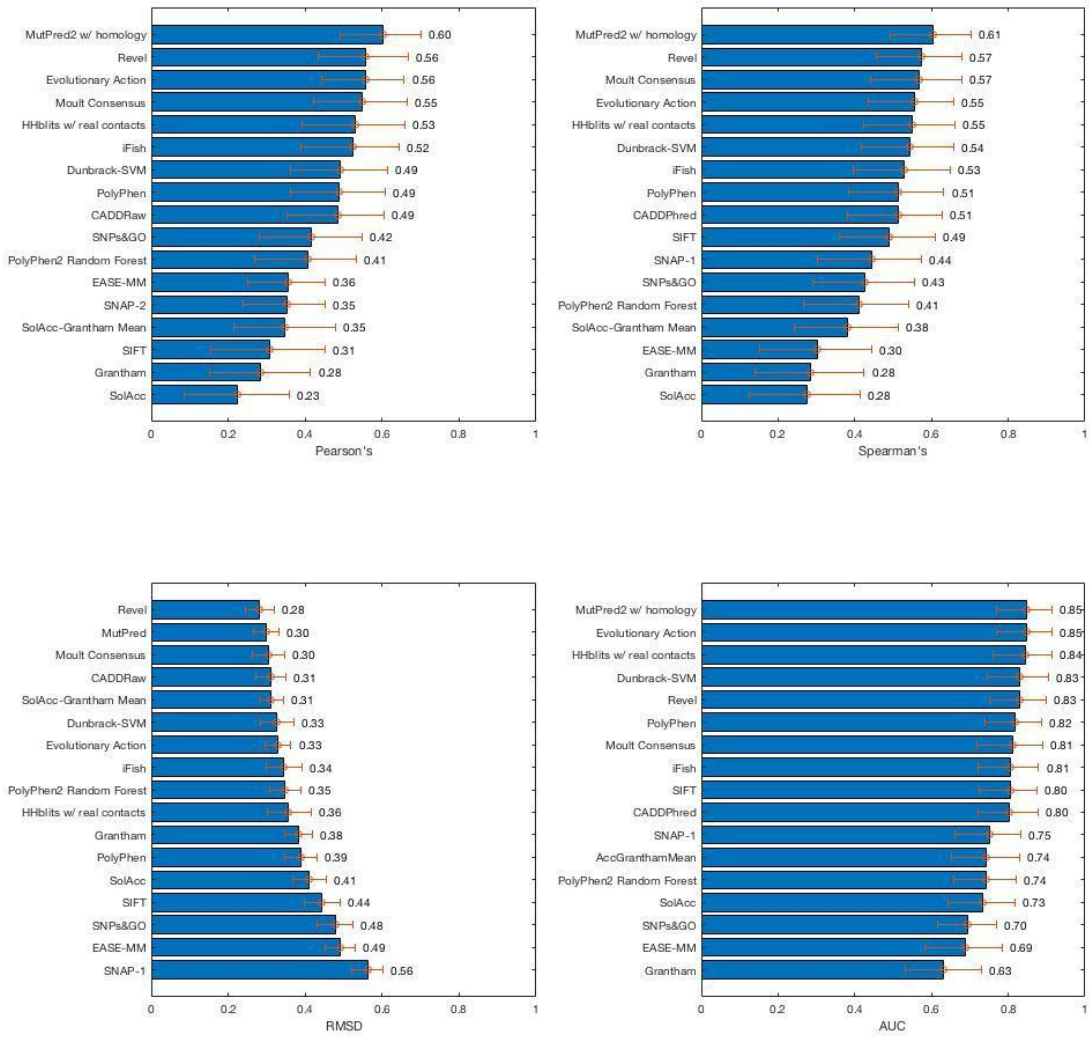
Supplemental Figures



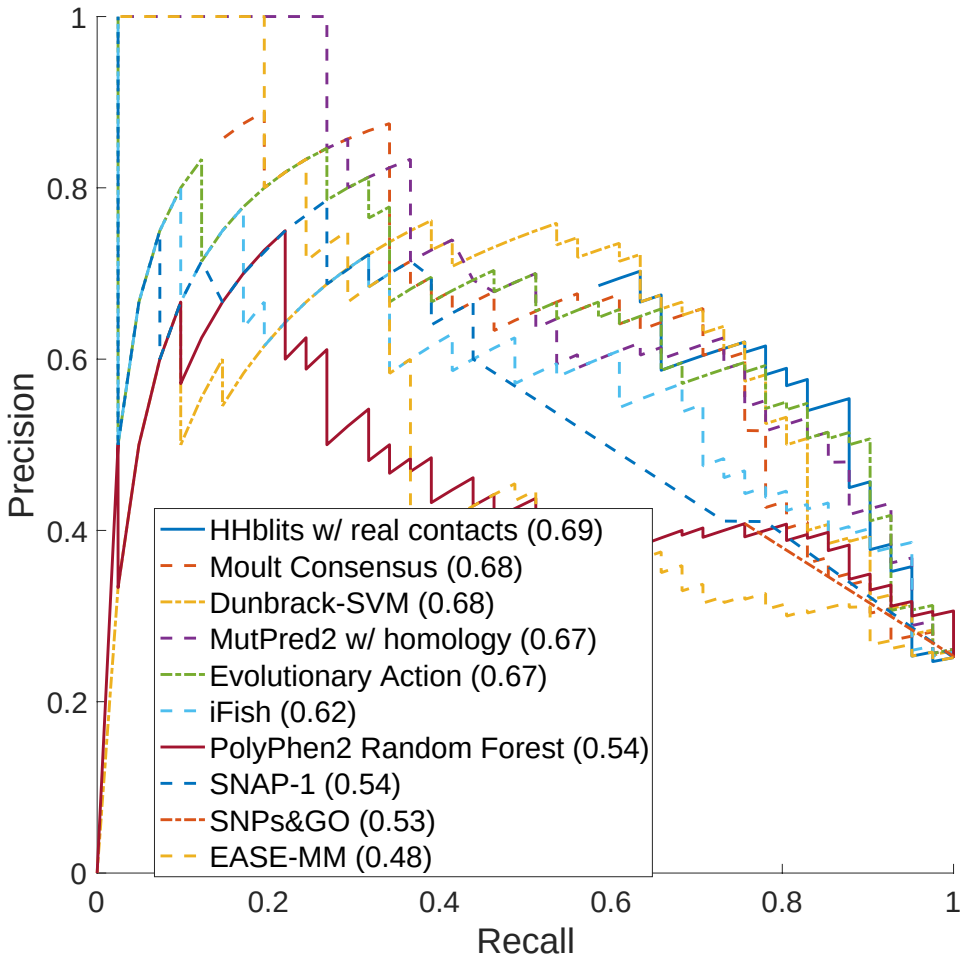
Supp. Figure S1: Amino acid positions and activity of tested variants

A plot showing the codon position and observed activity of tested variants. Variants with less than 15% activity are shown in red. A horizontal line designating the 15% activity mark is shown in grey.



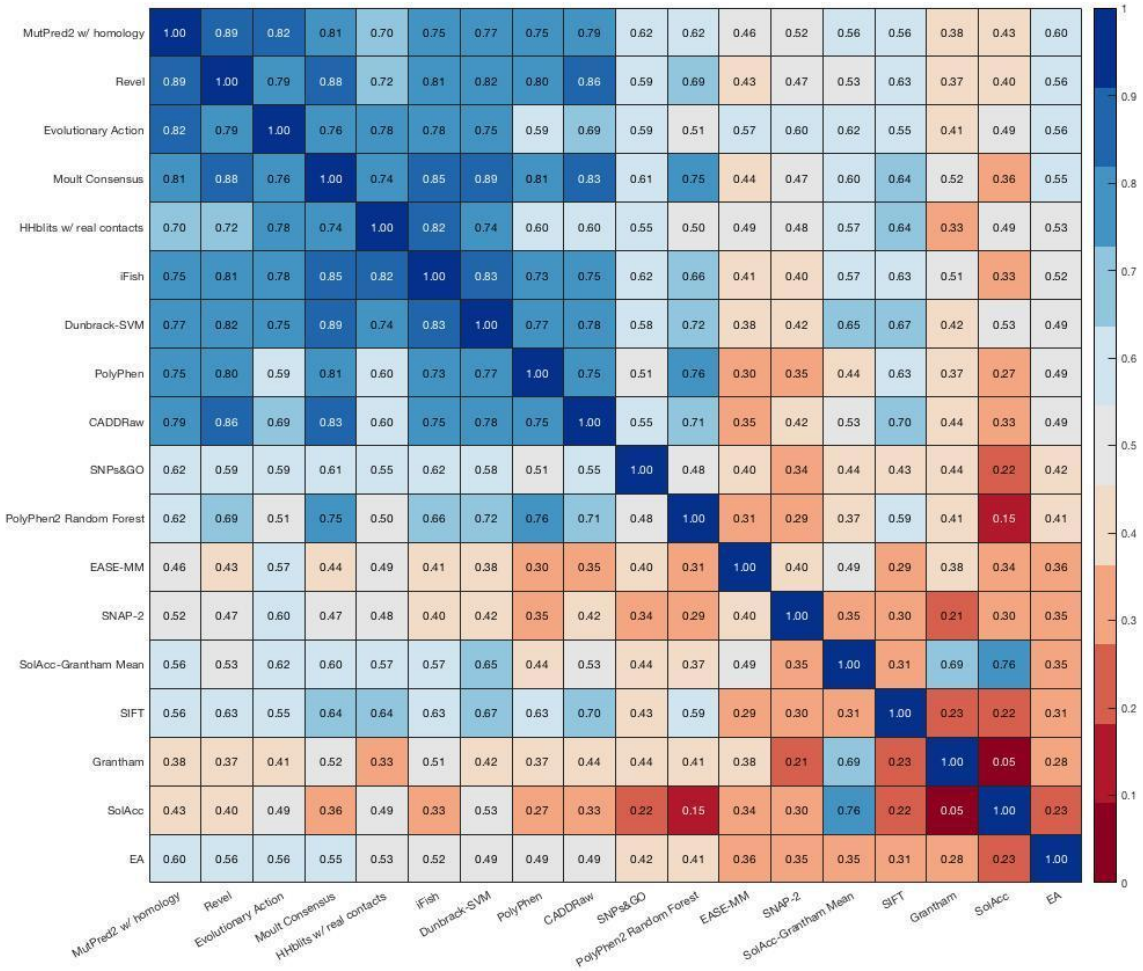
Supp. Figure S2: Evaluation Metrics

Resulting evaluation metrics for each submitted model in addition to several off the shelf tools (A: Pearson's r, B: Spearman's rho, C: RMSE, D: AUC). Error bars represent empirical 95% confidence intervals for each metric generated through 10^4 iterations of bootstrap sampling of data-points. Only the top performing model for each team is shown for each metric. AUC was calculated by designating missense mutations with less than 0.15 observed ***fWT*** activity as positives.



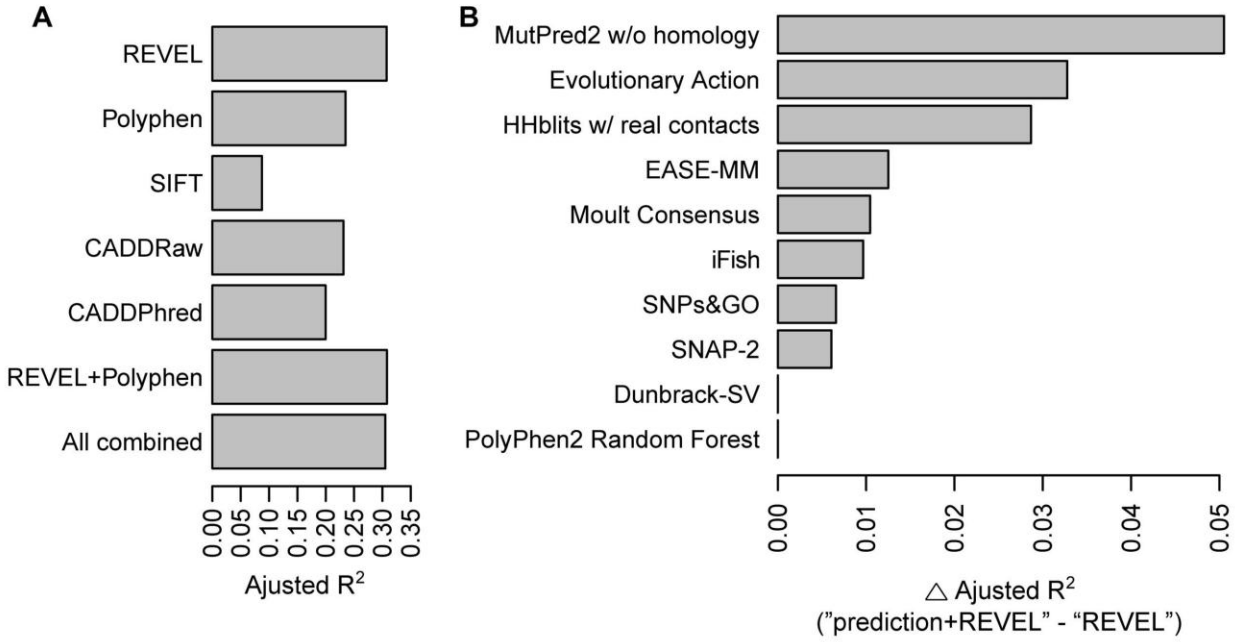
Supp. Figure S3: Precision and recall curves

Precision and recall curves for each team's top performing model when using 0.15 observed ***f*wt** activity as the threshold at which pathogenic and benign mutations are distinguished.



Supp. Figure S4: All vs All heatmap.

Pairwise Pearson's correlation coefficients between each team's top model as well as observed enzymatic activity values for submitted predictions as well as off the shelf methods.



Supp. Figure 5: Linear Mixed Model

Percentage of variance in predictions explained by each model including predictions from supplemental predictors.

Supp. Table S1: Pearson's correlation coefficients, Spearman's correlation coefficients, RMSE values and AUC values for all submitted models for all teams. Standard deviation values were calculated by resampling the data with replacement 10^4 times and re-calculating each respective metric.

Supp. Table S2: AUC values observed when using a range of observed ***fwt*** activity values to distinguish pathogenic and non-pathogenic variants.

Supp. Table S3: F-Max values observed when using a range of observed ***fwt*** activity values to distinguish pathogenic and non-pathogenic variants.

Supp. Table S4: Pairwise correlation coefficients when calculating Pearson's correlation coefficients between all considered models as well as ***f*wt** values when sampling variants with replacement 10^4 times.

Supp. Table S5: Pairwise standard deviations in correlation coefficients when calculating Pearson's correlation coefficients between all considered models as well as ***f*wt** values when sampling variants with replacement 10^4 times.

Supp. Table S6: P-values when calculating pairwise Pearson's correlation coefficients between all considered models as well as ***f*wt** values when sampling variants with replacement 10^4 times.

Supp. Table S7: R^2 values for each model.

Supp. Table S8: R^2 values observed when a multiple linear regression model was generated by combining the listed model with the best model (MutPred2 w/o Homology).

Supp. Table S9: Pearson's correlation coefficients, Spearman's correlation coefficients, RMSE values and AUC values for all submitted models for all teams as well as supplemental "off the shelf" methods. Standard deviation values were calculated by resampling the data with replacement 10^4 times and re-calculating each respective metric.

Supp. Table S10: Per-variant RMSE values when considering all models or the top 5 models (Mutpred, MutPred2 w/o homology, Moult consensus, Dunbrack-SVM, and Evolutionary Action)

Supplemental Model Descriptions: A zip file of model descriptions submitted by participating teams.